



Cyber Attacks Explained

Packet Crafting

Part—6

Protect your FOSS-based IT infrastructure from packet crafting by learning more about it.

In the previous articles in this series, we explored common infrastructure vulnerability exploitation scenarios. This article takes a step forward to describe a technically advanced attack that could badly impact networks because it is tough to detect. Packet crafting is a commonly used yet complex method to exploit vulnerabilities and, hence, it is important for network administrators to know more about it, and also to understand the various ways to protect their infrastructure against it. We will also look at how to protect FOSS-based systems from a packet crafting attack.

What is packet crafting?

Crafting, by definition, means to make or create something skillfully. As we know, all the vulnerability assessment tools used by network administrators to test

the security of their networks are both a blessing and a curse. This is because the same set of tools can also be used by evil hackers to find vulnerabilities and then exploit those to their benefit. Packet crafting, too, is not an exception to this rule, and since it is a technically advanced yet complex type of vulnerability exploitation, it is difficult to detect and diagnose.

Let's look at a TCP packet and its fields in detail, in order to understand a packet crafting attack better. Please refer to Figure 1, which shows a basic Ethernet packet as well as the TCPIP packet frame that rides on top of it. The Ethernet frame contains multiple fields that typically take care of Layer-2 communication, whereas the TCP and IP packet chunks carry data fields for the upper layers. The TCP part of the packet ensures a successful transport, while the IP packet